

Python Assignment – Week 13

Objective:

This assignment focuses on basic mathematical calculations and logic building using Python. You will write Python programs to perform temperature conversion, BMI calculation, and solve two additional problems: area calculation and number series analysis.

Problems:

1. Temperature Conversion (10 Points)

Problem Statement:

Write a Python program to convert a given temperature in **Celsius** to **Fahrenheit**.

Formula:

$$\text{Fahrenheit} = (\text{Celsius} \times 9/5) + 32$$

Example Input & Output:

```
Enter temperature in Celsius: 25
Temperature in Fahrenheit: 77.0
```

2. BMI Calculator (10 Points)

Problem Statement:

Write a Python program to calculate the **Body Mass Index (BMI)** using the formula:

Formula:

$$\text{BMI} = \frac{\text{Weight(kg)}}{\text{Height(m)}^2}$$

Example Input & Output:

```
Enter weight in kg: 70
Enter height in meters: 1.75
Your BMI is: 22.86
```

3. Area of a Circle (15 Points)

Problem Statement:

Write a Python program to calculate the **area of a circle** when the user provides the radius.

Formula:

$$\text{Area}=\pi\times(\text{radius})^2$$

(Use `math.pi` for the value of π)

Example Input & Output:

```
Enter the radius of the circle: 5
The area of the circle is: 78.54
```

4. Sum of Even and Odd Numbers (15 Points)**Problem Statement:**

Write a Python program to find the **sum of even and odd numbers separately** from a given range.

Example Input & Output:

```
Enter the start of range: 1
Enter the end of range: 10
Sum of even numbers: 30
Sum of odd numbers: 25
```